

Excel Formula Practice Laboratory: 30 Additional Exercises

Target Environment: excel_student_practice_250.csv (Columns A to I)

Instruction Set: This question bank presents 30 entirely unique problems covering specialized logical intersections, sorting assistance, advanced statistics, and text structures. Solve these using cell calculations starting in row 2 of your workspace.

Part 1: Advanced Statistical Analysis & Variations

Q21 Standard Deviation of Corporate Pricing (STDEV.S)

Measure the market volatility or spread of the Unit_Price column values to understand retail variations.

```
=STDEV.S(G2:G251)
```

Q22 Determining Median Unit Costs (MEDIAN)

Find the exact middle price point of all products sold, which divides the higher half from the lower half of prices.

```
=MEDIAN(G2:G251)
```

Q23 Most Recurring Order Volume Value (MODE.SNGL)

Find out which specific product quantity level appears most frequently inside transaction column F.

```
=MODE.SNGL(F2:F251)
```

Q24 Identifying Lower Price Thresholds (PERCENTILE.INC)

Isolate the 25th percentile value for Unit_Price to establish the entry cost boundary for the bottom quarter of products.

```
=PERCENTILE.INC(G2:G251, 0.25)
```

Q25 Determining High-Tier Volume Cutoffs (PERCENTILE.INC)

Isolate the 90th percentile value of Total_Revenue (Column I) to figure out the minimum revenue threshold required to be considered a top-tier order.

```
=PERCENTILE.INC(I2:I251, 0.90)
```

Q26 Numerical Ranking Assessment (RANK.EQ)

In column L, determine the numeric standing of the first order's Total_Revenue relative to all 250 orders in the system.

```
=RANK.EQ(I2, $I$2:$I$251)
```

Part 2: Complex Boolean Combinations (AND, OR, NOT)

Q27 Strict Multi-Condition Verification (AND)

In an empty tracking column, write an expression that evaluates to TRUE only if the transaction belongs to the **North** region AND the Quantity is exactly **5**. Otherwise, return FALSE.

```
=AND(C2="North", F2=5)
```

Q28 Flexible Alternative Logical Check (OR)

Write an evaluation query returning TRUE if an order's payment method matches either **Apple Pay** OR **PayPal**.

```
=OR(H2="Apple Pay", H2="PayPal")
```

Q29 Exclusion Target Matching (NOT)

Check if an order came from anywhere *except* the **South** region using a strict logical exclusion function.

```
=NOT(C2="South")
```

Q30 Complex Strategic Filtering Flag (IF with AND)

If an order belongs to the **Electronics** category AND generates a **Total_Revenue** over \$150, flag it as **"High Priority"**. Otherwise, flag it as **"Normal"**.

```
=IF(AND(D2="Electronics", I2>150), "High Priority", "Normal")
```

Q31 Promotional Discount Rule Application (IF with OR)

Calculate a promotional credit: If the order region is **West** OR the payment method used is **Credit Card**, offer a 5% discount on total revenue. Otherwise, give 0.

```
=IF(OR(C2="West", H2="Credit Card"), I2*0.05, 0)
```

Part 3: Advanced Lookups & Position Evaluation

Q32 Isolating Index Placements (MATCH)

Locate the exact relative row index position inside the Order list range where code **"ORD-1025"** is positioned.

```
=MATCH("ORD-1025", A2:A251, 0)
```

Q33 Dynamic Row Index Extraction (INDEX)

Using a fixed row reference, retrieve whichever product label text is typed natively in the exact 45th index cell slot of the Product range.

```
=INDEX(E2:E251, 45)
```

Q34 Lookup with Wildcard Parameters (VLOOKUP with Asterisk)

Find the unit price of the first item entry whose product title string starts explicitly with the keyword letters **"Wire"**.

```
=VLOOKUP("Wire*", E2:G251, 3, FALSE)
```

Q35 Reverse Column Lookup Execution (XLOOKUP / INDEX-MATCH)

Unlike VLOOKUP, extract data moving left: find the matching **Order_ID** for a product listing written in cell M2 by searching the product column and pulling from the ID column.

```
=XLOOKUP(M2, E2:E251, A2:A251)
```

Part 4: String Mutation, Space Trimming, & Length Audits

Q36 Character Length Extraction (LEN)

Count the exact total number of characters, spaces included, inside the first cell of the Product description column.

```
=LEN(E2)
```

Q37 Isolating Substring Length Blocks (MID)

Extract a middle segment from the Order_ID (Column A): extract just the 4 digits starting immediately after the 5th character place.

```
=MID(A2, 5, 4)
```

Q38 Annihilating Ghost Spacing Anomalies (TRIM)

Clean up text errors by stripping away any accidental double lead-in or trailing whitespace inputs from product descriptors.

```
=TRIM(E2)
```

Q39 Sentence Case Forcing Adjustments (PROPER)

Reformat category fields to ensure only the very first letter of every single word block reads as an uppercase capital letter.

```
=PROPER(D2)
```

Q40 String Subsegment Replacements (SUBSTITUTE)

Modify your data formatting display layout on the fly: change the structural hyphens inside your Order IDs to forward slashes instead (e.g. converting "ORD-1001" into "ORD/1001").

```
=SUBSTITUTE(A2, "-", "/")
```

Part 5: Chronological, Time, & Date Matrix Actions

Q41 Isolating Calendar Years (YEAR)

Extract just the 4-digit calendar year value out from the structured transaction Date entries listed in column B.

```
=YEAR(B2)
```

Q42 Isolating Calendar Months (MONTH)

Extract the numerical index number of the month (1 through 12) from the transaction date entry.

```
=MONTH(B2)
```

Q43 Determining Day Names (TEXT Function)

Convert the transaction date entry into a human-readable day name (e.g., turning a date into "Monday").

```
=TEXT(B2, "dddd")
```

Q44 Aging Duration Days Ledger (DAYS)

Determine the exact total number of calendar days that have elapsed between the transaction date in cell B2 and a fixed project deadline date of **December 31, 2026**.

```
=DAYS("2026-12-31", B2)
```

Part 6: Advanced Subtotaling & Smart Rounding Frameworks

Q45 Financial Rounding to Nearest Cent (ROUND)

Round a calculated complex percentage tax index figure to exactly 2 decimal decimal places.

```
=ROUND(I2 * 0.0825, 2)
```

Q46 Forced Round-Up Threshold Caps (ROUNDUP)

Ensure shipping calculations favor coverage margins by forcing a raw calculated weight value to always round UP to the nearest whole integer.

```
=ROUNDUP(G2 * 0.12, 0)
```

Q47 Truncating Values Natively (TRUNC)

Completely chop off and remove all decimal components from a revenue ratio cell without altering or rounding the base whole integer numbers.

```
=TRUNC(I2)
```

Q48 Filtered Rows Subtotal Counting (SUBTOTAL 3 - COUNTA alternative)

Write a smart counter function for Column A that updates automatically to show only the count of currently visible rows when user sorting filters are applied.

```
=SUBTOTAL(3, A2:A251)
```

Q49 Filtered Data Sales Aggregation (SUBTOTAL 9 - SUM alternative)

Calculate a dynamic total of your revenue column that recalculates automatically to sum up only the filtered rows currently showing on the screen.

```
=SUBTOTAL(9, I2:I251)
```

Q50 Array Empty-Cell Validation Checks (COUNTBLANK)

Audit the integrity of your data table: check across the entire database grid range to confirm there are zero empty rows or unpopulated data blocks.

```
=COUNTBLANK(A2:I251)
```